Code No: 111AE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, March/April - 2023
ENGINEERING CHEMISTRY
(Common to CE, EEE, ME, ECE, CSE, IT, MCT, AE, MIE, AGE)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define molar conductance and its units.	[2]
b)	Why moderate temperature is used during electroplating.	[3]
c)	Define tactility and functionality of a monomer.	[2]
d)	Mention the characteristics of a good lubricant.	[3]
e)	Write the structure for EDTA.	[2]
f)	Distinguish between temporary and permanent hardness.	[3]
g)	What is meant by ignition temperature?	[2]
h)	Why should fuel for cars not evaporate too easily?	[3]
i)	Name two set systems which form eutectic mixtures.	[2]
j)	Name the type of adsorption with examples.	[3]

PART - B

(50 Marks)

2.a)	Describe the construction of methanol-oxygen fuel cell with reactions.	
b)	Explain the mechanism of electro chemical corrosion.	[5+5]
	OR	
3.a)	Derive Nernst equation for single electrode potential. Mention its applications.	
b)	Write a brief note on sacrificial anodic impression.	[5+5]
4.a)	Explain in detail about injection molding method with a neat diagram.	
b)	Explain the preparation and applications of Butyl rubber and Thiokol rubber.	[5+5]
	OR	
5.a)	With the help of neat diagram write about thick film lubrication method.	
b)	Mention the applications of nanomaterials.	[5+5]
6.a)	Explain zeolite process for softening of water.	
b)	Write about reverse osmosis and its significance.	[5+5]
	OR	
7.a)	Discuss in detail about scale and sludge formation in water boilers. How is it controlled?	
b)	Write a brief notes on internal treatment of water.	[5+5]

8.a) Describe the analysis of flue gas by orsat's apparatus.

b) Define calorific value. Distinguish between the gross and net calorific value. [5+5]

OR

9.a) Define knocking of a fuel. Discuss in detail how to improve the knocking characteristics of a fuel.

b) Explain the fixed –bed catalytic method to obtain gasoline. Give its mechanism. [5+5]

10.a) Draw a labelled diagram of water system and discuss the meta stable curve and principle of freeze drying.

b) Mention the classification of colloids with examples. [5+5]

OR

11.a) State Phase rule and discuss its applications to water system.

b) Distinguish between absorption and adsorption. [5+5]

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